👩‍🏫 Staff Guide · Horticulture 1 · Unit 8 Poinsettias (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Her readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as her worksheets for that lesson
- Quick check: score, then scale to her test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Poinsettia Final Test**: Paper final not in the export; use our quiz.
- **Common Poinsettia Pests**: Assigned in class; not in the export.
- **Smith Gardens field trip**: Signed travel form + medical info required (in the syllabus packet).

💗 Note

Poinsettia sap: latex-family irritant. Check health plans for latex allergy; use nitrile gloves. Students never handle pesticides or growth regulators.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

🌺 Poinsettias: Guided Notes

Horticulture 1 · Unit 8 · Level 3 (closer to the class work) · goes with Unit 13 Questions, care chart, pests, lab report, Poinsettia Portfolio, quizzes

Name: _____

Use the class slides and your reading. Key words:

poinsettia *bract* *cyathia* *photoperiod* *short-day plant* *botrytis* *root rot* *whitefly* *sap* *sleeve*

A. Basics

1. Poinsettias are native to _____.
2. The colored parts are _____; the true flowers are _____.

B. Photoperiod

3. Explain how a grower makes poinsettias turn red by December.

_____

_____

C. Care

4. Fill in the table.

Problem	Cause	Prevention
Botrytis		
Root rot		
Whiteflies		

🌺 Poinsettias: Reading

Horticulture 1 · Unit 8 · Level 3 (closer to the class work) · goes with Unit 13 Questions, care chart, pests, lab report, Poinsettia Portfolio, quizzes

Name: _____

Growing Poinsettias

Poinsettias are native to Mexico and were brought to the United States in the 1820s by Joel Poinsett. Today they are the top-selling potted flowering plant in the country, with most sold in November and December. Growers start them from cuttings in summer so every plant in a crop is identical.

The colorful parts of a poinsettia are not petals. They are bracts, modified leaves that turn red, pink, white or marbled. The true flowers, called cyathia, are the small yellow structures in the center. Buyers look for cyathia that have not yet shed pollen, a sign that the plant is fresh.

Poinsettias are short-day plants. Their bracts color only when they receive long, uninterrupted nights of about 14 hours of darkness for several weeks, starting in late September or early October. Even light from a street lamp can keep them green. Photoperiod is the length of the light period in a day.

Good care keeps poinsettias healthy. Water the soil when the surface feels dry and keep water off the leaves, because wet foliage invites botrytis, a gray mold. Overwatering causes root rot. Common pests include whiteflies, mealybugs and mites. The milky sap can irritate skin and eyes and is a concern for people with latex allergies, so wear nitrile gloves. Pesticides and growth regulators are applied only by staff. At the sale, plants are sleeved to protect them from cold and damage.

Check your reading

1. Explain why a poinsettia's red parts are not flowers.

2. What does short-day plant mean for a grower?

3. Name two care mistakes and the problem each causes.

🌺 Poinsettia Care & Pinching: Lab Sheet

Horticulture 1 · Unit 8 · Level 3 · goes with Care chart + Pinching Plants + Growing Instructions

Name: _____

Question: How does our care affect poinsettia growth and coloring over time?

Materials

poinsettia, care chart, ruler, hand lens

⚠️ Safety

- Nitrile gloves (sap/latex)
- Students never handle pesticides
- Wash hands

Prediction

How many weeks until the bracts are fully colored?

_____

_____

Procedure

1. Check moisture; water soil if dry.
2. Inspect for pests.
3. Measure height and count colored bracts weekly.
4. Log on the care chart.

Data

Week	Height (cm)	Bracts colored	Action / observation
1			
2			
3			
4			

Analysis (Claim, Evidence, Reasoning)

1. Claim: is your plant on track for the sale? Evidence from your chart; reasoning about photoperiod and care.

🔬 Poinsettia 3-Plant Experiment: Lab Report Frame

Horticulture 1 · Unit 8 · Level 3 · goes with “How to Write a Scientific Lab Report”

Name: _____

Title and Introduction

State the question, background on poinsettias (photoperiod, care), and your hypothesis (If… then… because…).

Materials and Method

List materials. Describe the variable you changed, the control plant, and what you measured and how often.

Results and Graph

Summarize the data with numbers and include a labeled graph (title, axes, units).

Discussion

Explain why the results happened using what you know about poinsettias. Note errors or problems.

Conclusion

Was the hypothesis supported? What would you test next?

🌺 Poinsettias: Quick Check

Horticulture 1 · Unit 8 · Level 3 (closer to the class work) · goes with Unit 13 Questions, care chart, pests, lab report, Poinsettia Portfolio, quizzes

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Poinsettias get their color in the…

A. Flowers | B. Roots | C. Bracts

2. To change color, poinsettias need…

A. Short days and long nights | B. Long days | C. More fertilizer

3. Overwatering causes…

A. Botrytis only | B. Red bracts | C. Root rot

4. Poinsettias are native to…

A. Mexico | B. Peru | C. Spain

5. Why should you not get water on poinsettia leaves?

6. Why is a poinsettia sleeved when sold?

🌺 Poinsettias: Guided Notes · ANSWER KEY

Horticulture 1 · Unit 8 · Level 3 (closer to the class work) · goes with Unit 13 Questions, care chart, pests, lab report, Poinsettia Portfolio, quizzes

Name: _____

Use the class slides and your reading. Key words:

poinsettia · *bract* · *cyathia* · *photoperiod* · *short-day plant* · *botrytis* · *root rot* · *whitefly* · *sap* · *sleeve*

A. Basics

1. Poinsettias are native to **Mexico**.
2. The colored parts are **bracts**; the true flowers are **cyathia**.

B. Photoperiod

3. Explain how a grower makes poinsettias turn red by December.
Give ~14 hours of total darkness each night from early October for 8-10 weeks.

C. Care

4. Fill in the table.

Problem	Cause	Prevention
Botrytis	**wet leaves**	**water soil, air flow**
Root rot	**overwatering**	**water when surface dry**
Whiteflies	**insects**	**inspect under leaves; staff treat**

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🌺 Poinsettia Care & Pinching: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 8 · Level 3 · goes with Care chart + Pinching Plants + Growing Instructions

Name: _____

Question: How does our care affect poinsettia growth and coloring over time?

Materials

poinsettia, care chart, ruler, hand lens

⚠️ Safety

- Nitrile gloves (sap/latex)
- Students never handle pesticides
- Wash hands

Prediction

How many weeks until the bracts are fully colored?
About 8-10 weeks from the start of dark treatment.

Procedure

1. Check moisture; water soil if dry.
2. Inspect for pests.
3. Measure height and count colored bracts weekly.
4. Log on the care chart.

Data

Week	Height (cm)	Bracts colored	Action / observation
1	**20**	**0**	**watered, pinched**
2	**22**	**0**	**no pests**
3	**24**	**2**	**watered**
4	**25**	**5**	**whitefly reported**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: is your plant on track for the sale? Evidence from your chart; reasoning about photoperiod and care.
Answers vary; coloring should increase with long nights.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🌺 Poinsettias: Quick Check · ANSWER KEY

Horticulture 1 · Unit 8 · Level 3 (closer to the class work) · goes with Unit 13 Questions, care chart, pests, lab report, Poinsettia Portfolio, quizzes

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Poinsettias get their color in the…

A. Flowers	B. Roots	**C. Bracts**

2. To change color, poinsettias need…

A. Short days and long nights	B. Long days	C. More fertilizer

3. Overwatering causes…

A. Botrytis only	B. Red bracts	**C. Root rot**

4. Poinsettias are native to…

A. Mexico	B. Peru	C. Spain

5. Why should you not get water on poinsettia leaves?

Wet leaves can cause botrytis and damage.

6. Why is a poinsettia sleeved when sold?

To protect it from cold and damage in transport.

🔑 *Reworded from her Poinsettia Test and Final Quiz (keys in the export), please check.*